MEMORANDUM: Q1 Facility Optimization & Penalty Governance

STANDARDS: Compliant With US ANSI/IISE Manufacturing Metrics

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1. Introduction

"If you can not measure it, you can not improve it."(Lord Kelvin)

It is widely believed that improving productivity is currently the biggest challenge for companies seeking to remain competitive in a global market [1, 2]. A well-known way to measure effectiveness is the Overall Equipment Efficiency (OEE) index. It was first developed by the Japan Institute for Plant Maintenance (JIPM), and it is widely used in many industries. Moreover, it is the backbone of quality improvement methodologies, such as TQM and Lean Production.

The strength of the OEE index lies in making losses more transparent and highlighting areas for improvement. OEE is often seen as a catalyst for change, and it is easy to understand, as a lot of articles and discussions have been generated about this topic over the last few years.

The aim of this chapter is to answer the general questions of what to measure. *how to measure?* And how to use the measurements to optimize factory performance? The goal is to show that OEE is a good base for optimizing the factory performance. Moreover, OEE's evolutions are the perfect response even in advanced frameworks.

This chapter begins with an explanation of the difference between efficiency, effectiveness and productivity as well as with a formal definition for the components of effectiveness. Mathematical formulas for calculating OEE are provided too.

After the introduction to the fundamental of OEE, some interesting issues concerning the way to implement the index are investigated. Starting with the question that in calculating OEE you have to take into consideration machines as operating in a linked and complex environment. So we analyze almost a model for the OEE calculation that lets a wider approach to the performance of the whole factory. The second issue concerns with monitoring the factory performance through OEE. It implies that information for decision-making have to be guaranteed real-time. It is possible only through automated systems for calculating OEE and through the capability to collect a large amount of data. So we propose an examination of the main automated OEE systems from the simplest to high-level systems integrated into ERP software. Even data collection strategies are screened for rigorous measurement of OEE.

The last issue deals with how OEE has evolved into tools like TEEP, PEE, OFE, OPE and OAE in order to fit with different requirements.

At the end of the chapter, industrial examples of OEE application are presented and the results are discussed.

2. Fundamentals of OEE

Overall equipment efficiency or effectiveness (OEE) is a hierarchy of metrics proposed by Seiichi Nakajima [3] to measure the performance of the equipment in a factory. OEE is a really powerful tool that can be used also to perform diagnostics as well as to compare production units in differing industries. The OEE has born as the backbone of Total Productive Maintenance (TPM) and then of other techniques employed in asset management programs, Lean manufacturing [4], Six Sigma [5], World Class Manufacturing [4].

By the end of the 1980's, the concept of Total Production Maintenance became more widely known in the Western world [7] and along with it OEE implementation too. From then on an extensive literature [8-11] made OEE accessible and feasible for many Western companies.

3. Difference between efficiency, effectiveness and productivity

Confusion exists as to whether OEE has indeed been an effectiveness or efficiency measure. The traditional vision of TMP referred to Overall Equipment Efficiency while now it is generally recognized as Overall Equipment Effectiveness. The difference between efficiency and effectiveness is that effectiveness is the actual output over the reference output and efficiency is the actual input over the reference input. The Equipment Efficiency refers thus to ability to perform well at the lowest overall cost. Equipment Efficiency is then unlinked from output and company goals. Hence the concept of Equipment Effectiveness relates to the ability of producing repeatedly what is intended producing, that is to say to produce value for the company (see Figure 1).

Productivity is defined as the actual output over the actual input (e.g. number of final products per employee), and both the effectiveness and the efficiency can influence it. Regarding to OEE, in a modern, customer-driven "lean" environment it is more useful to cope with effectiveness.

4. Formal definition of OEE

According to the previous remark a basic definition of OEE is:

$$\frac{ValuableOperatingTime}{LoadingTime}$$

where:

- Valuable Operating Time is the net time during which the equipment actually produces an acceptable product;
- Loading Time is the actual number of hours that the equipment is expected to work in a specific period (year, month, week, or day). [1]

The formula indicates how much the equipment is doing what it is supposed to do and it captures the degree of conforming to output requirements. It is clearly a measure of effectiveness.

OEE is not only a metric, but it also provides a framework to improve the process. A model for OEE calculation aims to point out each aspect of the process that can be ranked for improvement. To maximize equipment effectiveness it is necessary to bring the equipment to peak operating conditions and then keeping it there by eliminating or at least minimizing any factor that might diminish its performance. In other words a model for OEE calculation should be based on the identification of any losses that prevent equipment from achieving its maximum effectiveness.

The OEE calculation model is then designed to isolate losses that degrade the equipment effectiveness.

5. Losses analysis

Losses are activities that absorb resources without creating value. Losses can be divided by their frequency of occurrence, their cause and by different types they are. The latter one has been developed by Nakajima [4] and it is the well-known Six Big Losses framework. The other ones are interesting in order to rank rightly losses.

Losses can be chronic or sporadic. The chronic disturbances are usually described as *"small, hidden and complicated"* while the sporadic ones occur quickly and with large deviations from the normal value. The loss frequency combined with the loss severity gives a measure of the damage and it is useful in order to establish the order in which the losses have to be removed. This classification makes it possible to rank the losses and remove them on the basis of their seriousness or impact on the organization.

Regarding divide losses by their causes, three different ones can be found:

1. machine malfunctioning: an equipment or a part of this does not fulfill the demands;
2. process: the way the equipment is used during production;
3. external: cause of losses that cannot be improved by the maintenance or production team.

The external causes such as shortage of raw materials, lack of personnel or limited demand do not touch the equipment effectiveness. They are of great importance for top management and they should be examined carefully because their reduction can directly increase the revenues and profit. However they are not responsible of the production or maintenance team and so they are not taken into consideration through the OEE metric.

To improve the equipment effectiveness the losses because of external causes have to be taken out and the losses caused by machine malfunctioning and process, changeable by the daily organization, can still be divided into:

Down time losses: when the machine should run, but it stands still. Most common downtime losses happen when a malfunction arises, an unplanned maintenance task must be done in addition to the big revisions or a set-up/start-up time occurs.

Speed losses: the equipment is running, but it is not running at its maximum designed speed. Most common speed losses happen when equipment speed decrease but it is not zero. It can depend on a malfunctioning, a small technical imperfections, like stuck packaging or because of the start-up of the equipment related to a maintenance task, a setup or a stop for organizational reasons.

Quality losses: the equipment is producing products that do not fully meet the specified quality requirements. Most common quality losses occur because equipment, in the time

between start-up and completely stable throughput, yields products that do not conform to quality demand or not completely. They even happen because an incorrect functioning of the machine or because process parameters are not tuned to standard.

Equipment Defect Penalties

To maintain rigorous OEE standards, the Operations contractor is penalized for poor machine reliability.

- The Standard Unplanned Equipment Defect Penalty is calculated by multiplying the Total Defect Downtime Minutes by $800.00. [2]
- The resulting dollar amount is charged directly to the Maintenance Department's quarterly budget

6. Attacking the six big losses

By having a structured framework based on the Six Big Losses, OEE lets to track underlying issues and root causes. By knowing what the Six Big Losses are and some of the causes that contribute to them, the next step is to focus on ways to monitor and correct them. In the following let's see what is the way:

Breakdown: eliminating unplanned downtime is critical to improving OEE. Other OEE factors cannot be addressed if the process is down. It is not only important to know how much and when down time equipment is but also to be able to link the lost time to the specific source or reason for the loss. With down time data tabulated, the most common approach is the Root Cause Analysis. It is applied starting with the most severe loss categories.

Set-up and adjustments: tracking setup time is critical to reducing this loss. The most common approach to reduce this time is the Single Minute Exchange of Dies program (SMED).

Minor stoppages and Reduced speed: minor stoppages and reduced speed are the most difficult of the Six Big Losses to monitor and record. Cycle Time analysis should be utilized to point out these loss types. In most processes recording data for Cycle Time analysis needs to be automated since the cycles are as quick as they do not leave adequate time for manual data logging. By comparing all cycles to the theoretical Cycle Time, the losses can be automatically clustered for analysis. It is important to analyze Minor stoppages and Reduced speed separately because the root causes are typically very different.

Quality losses and Reduced yield: parts that require rework of any kind should be considered rejects. Tracking when rejects occur and the type is critical to point out potential causes, and in many cases patterns will be discovered. Often a Six Sigma program, where a common metric is achieving a defect rate of less than 3.4 defects per million opportunities, is used to focus attention on a goal of "zero defects".

7. OEE evolution: TEEP, PEE, OAE, OFE, and OPE

During the last decades, both practitioners and researchers have raised the discussion about OEE in many ways. One of the most popular has led to modification and enlargement of individual original OEE tool to fit a broader perspective as supposed important for the companies [16]. With the evolution of OEE, different definitions have also come up in literature and in practice, coupled with their changed formulations. Some of these formulations (TEEP and PEE) are still at the equipment level, while the others (OAE, OFE and OPE) extended OEE to the factory level. Let's go through the main features of each formulation.

TEEP stands for Total Equipment Effectiveness Performance and it was proposed firstly by Invancic [17]. TEEP is a performance metric that shows the total performance of equipment based on the amount of time the equipment was present. So OEE quantifies how well a manufacturing unit performs relative to its designed capacity, during the periods when it is scheduled to run. TEEP measures OEE effectiveness against Calendar Time, i.e.: 24 hours per day, 365 days per year.

8. OEE for the factory

As mentioned in the previous section equipment operates in a linked and complex environment. So it is necessary to pay attention beyond the performance of individual tools towards the performance of the whole factory. According to Scott and Pisa [21], the answer to this requirement is the OFE metric, which is about combining activities and relationships between different parts of the equipment, and integrating information, decisions, and actions across many independent systems and subsystems. The problem is that a specific and unique method to calculate OFE does not exist. There many methodologies and approaches, with different level of complexity, different information coming from and different lacks.

9. What is OEE for?

OEE provides simple and consolidated formulas to measure effectiveness of the equipment or production system. Moreover Dal et al. [31] point out that it can also be used as an indicator of process improvement activities since OEE is directly linked to the losses as well as OEE can be even used to compare performance across the factory highlighting poor line performance or to quantify improvements made [31]. Moreover improving can be pursued by:

- Backtracking to determine what loss reduces effectiveness.
- Identifying bottlenecks as not only the slowest machine, but as the machine both slower and less effective.

10. Data collection strategies

The OEE calculations should be based on correct input parameters from the production system as reported by Ericsson [33]. Data acquisition strategies range from very manual to very automated. The manual data collection method consists of a paper template, where the operators fill in the cause and duration of a breakdown and provide comments about minor stoppages and speed losses. It is a low-tech approach. On the contrary a high-tech approach runs through an automatic OEE calculation system that is governed by sensors connected to the equipment, automatically registering the start time and duration of a stoppage and prompting the operator to provide the system with information about the downtime cause. An automatic approach usually provides opportunities to set up lists of downtime causes, scheduling the available operating time and making an automatic OEE calculation for a time period. A variety of reports of production performance and visualization of the performance results are even possible to retrieve from the system.

11. Automating OEE and integration of OEE into automated management system

Automating OEE gives a company the ability to collect and classify data from the shop floor into meaningful information that can help managers understand the root causes of production inefficiency. Therefore giving greater visibility to make more informed decisions on process improvement. An automated OEE system addresses the three primary functions of OEE:

Acquisition: it concerns data collection that as discussed above data will be completely automatic.

Analysis: it usually provides algorithms to calculate OEE and other items related to. Moreover it is often able to support downtime classification via reason trees and other technical analysis. The more sophisticated the package, the more analysis equipment is available.

Visualization: OEE metrics are available through reports or they can be displayed even via a software interface directly to the operator.

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[2] Process & Layout Bottlenecks (Exception Clause)

Industrial Engineering principles dictate that machines should not overproduce.

- If a machine enters an idle or fault state strictly triggered by downstream Work-In-Progress (WIP) buffer limits (e.g., thermal cooling constraints), the machine is functioning as designed under lean pull-system logic.

- These events are classified as "Process/Layout Bottlenecks," rather than "Equipment Defects."

- Process/Layout Bottlenecks carry a $0 Maintenance Penalty. Instead, a flat $400,000 "Line Rebalancing Assessment" must be assessed against the internal Industrial Engineering budget to fund facility redesign.

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